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How Can Chatbots Comfort Humans? Exploring the Effects of Emoji and Identity Cues in Emotional Support

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ABSTRACT
Based on the Computers Are Social Actors (CASA) paradigm, this study examines how the identity cues (human vs. chatbot) and the use of emoji influence the effectiveness of chatbots in providing emotional support in the context of Hong Kong. Through a 2 x 2 online experiment with 494 valid participants, the results showed that the interaction between emoji use and identity cues significantly affected the social presence. Only in the condition of human identity cues, using emojis in the message could significantly increase social presence, which enhanced participants' perceived warmth and perceived competence regarding the communication partner. Moreover, these social perceptions could further increase perceived emotional support when human identity cues were present. Theoretically, this study extends the CASA paradigm into the domain of supportive communication by identifying a perception-cognition-behavior pathway driven by human-like social cues. Practically, it highlights the context-dependent nature of multiple social cues in effective human-computer interaction, offering insights for culturally adaptive chatbot design.

KEYWORDS
Supportive chatbots; identity cues; emojis; social presence; CASA paradigm

1. Introduction

Chatbots have been developed in various fields of application, ranging from customer services (Følstad & Skjuve, 2019), entertainment (Jin et al., 2019), and education (Neto & Fernandes, 2019) to mental health (Miner et al., 2017). As a new supportive communication partner in different fields, one of the critical communication functions of chatbots is to provide social support (Meng et al., 2023). Specifically, chatbots have been increasingly used to provide emotional support to people undergoing stress. For instance, a group of psychologists and AI (Artificial Intelligence) experts at Stanford University have developed a chatbot named "Woebot" on a cognitive-behavioral therapy (CBT) platform to help people manage their mental health conditions by enabling them to reconstruct their negative thoughts into positive ones. Replika is another U.S. chatbot product that aims to build emotional connections with its users and provides customized chatting services to customers based on their conversations. As previous systematic reviews of AI and health care have summarized, the application of chatbots predominates in the mental health field (Laranjo et al., 2018). It has been a cost-efficient way of providing emotional support, especially during the COVID-19 pandemic and beyond.

Existing research shows that users often respond to chatbots based on human-like social cues, consistent with the CASA paradigm (Nass et al., 1994). Prior work demonstrates that users express politeness toward computers (Adam et al., 2021), apply gender stereotypes through design elements such as

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color (Carolus et al., 2018), and interpret emojis as social signals that encourage them to treat machines as social actors (Fogg, 2003) or as more humanlike (Beattie et al., 2020). These findings suggest that social communication cues embedded in chatbot design can shape user perceptions, psychological reactions, and the broader human–machine interaction process. Recent studies have further examined emotional communication styles in chatbots. For example, emoji use can lower usage intention depending on the chatbot's personality and the context of the task (Kuhail et al., 2025). Humorous emojis can increase reuse intention after service failure by enhancing perceived intelligence (Liu et al., 2023). Additionally, employing human-like language can improve enjoyment and social presence, even when chatbots make mistakes (de Sá Siqueira et al., 2024). Together, this body of work indicates that emotional cues influence user responses in nuanced and context-sensitive ways.

Despite the growing interest in supportive chatbots and user perception, several critical research gaps persist. First, although recent studies have examined emoji use in various chatbot contexts, their roles as nonverbal cues in emotionally supportive interactions is still not well understood. Second, there is a limited understanding of the psychological mechanisms through which users form evaluations of emotional support. Third, while some studies have begun incorporating real interactive chatbot systems (Meng & Dai, 2021), many still rely on static scripts method (e.g., Merrill et al., 2022; Yu & Zhao, 2024), which limits ecological validity. Furthermore, existing research in this domain remains predominantly Western-centric (e.g., Beattie et al., 2020; Rheu et al., 2024), which may not fully capture how users in different environments interpret nonverbal cues.

As noted above, the effects of emojis in chatbot communication are context-sensitive. This sensitivity may be particularly relevant in high-context communication environments, such as those commonly described in Chinese cultural settings (Bello et al., 2010), where interpersonal meaning relies more heavily on implicit and nonverbal cues. In such environments, emojis may serve as particularly salient paralinguistic signals in supportive interactions.

Hong Kong, located in the Greater Bay Area of China, provides a relevant setting for examining these issues not only because it is embedded in a high-context cultural environment but also because it has actively promoted digital mental health technologies. Local initiatives such as MindHK's "Help Me Virtual Assistant" and the "Lihkg Cantonese Bot" illustrate a growing technological infrastructure and public openness toward conversational support tools. This context provides a practical setting for examining the role of chatbot design in emotional support. In view of these contextual considerations, we propose the following research question:

> How does the design of chatbots (identity cues and emoji adoption) influence the effectiveness of emotional support?

To address this research question, this study examines how two common chatbot design features—identity cues and emoji use shape users' perceptions of emotional support by enhancing social presence and emotional connection, particularly in high-context cultural settings. Theoretically, it positions CASA within contemporary human-machine communication (HMC) perspectives and extends it to the emotional-support context. Practically, it offers implications for designing chatbots that foster emotional support through contextually adaptive social cues.

2. Literature review

2.1. Chatbots and emotional support

People may derive various emotional benefits from interacting with chatbots. A growing body of research has examined the relational experiences users develop through such interactions, including feelings of enjoyment, attachment, and relationship satisfaction. For example, users often report a sense of enjoyment when engaging with virtual agents (Heerink et al., 2009) and may even form feelings of friendship with them (Skjuve et al., 2021). These companionship-like experiences, facilitated by media technologies, can contribute to users' perceptions of emotional support (Ta et al., 2020).

Similar to support in human-human interaction, scholars believe that the objective of communicating with supportive chatbots is to obtain effective help (Beattie & High, 2022). Emotional support is

regarded as the most used type of support (Burleson, 2008) and applies to various stressor arrangements (MacGeorge et al., 2014). Emotional support, which can be defined as expressions of care, love, compassion, and empathy, is one type of social support (Cutrona & Suhr, 1992). The growing body of research on supportive chatbots has primarily focused on two aspects, namely how various design features facilitate the delivery of emotional support, and how users emotionally respond to such interactions (Liu et al., 2025).

In terms of design features, studies have highlighted the role of chatbot personalization and anthropomorphic cues. For instance, social chatbots like Replika, which incorporate personalized interaction patterns, have been shown to help users develop friendship-like emotional bonds (Brandtzaeg et al., 2022; Skjuve et al., 2021). More recent research indicates that the degree of anthropomorphism and perceived authenticity significantly shape users' emotional attachment (Pentina et al., 2023). Regarding users' emotional responses, evidence shows that emotional disclosure from the chatbots can enhance users' positive affect (Ho et al., 2018), and that reciprocal self-disclosure from chatbots can effectively reduce stress and anxiety (Meng & Dai, 2021). Supporting this line of work, Meng et al. (2023) further demonstrate that highly interactive chatbots can provide emotional support comparable to that of human supporters, particularly by fostering cognitive reappraisal.

Despite these advances, limited attention has been paid to how specific design cues, such as identity cues and emoji use, jointly shape users' perceptions of emotional support and the underlying psychological mechanisms. The information elements (especially the nonverbal aspects) that signal emotional intent and social presence remain underexplored. Building on this foundation, Rheu et al. (2024) emphasize the importance of managing users' expectations and refining feedback strategies to optimize supportive interactions. Given that emotional support is inherently conveyed through both verbal and nonverbal channels (Burleson, 2008; Cutrona & Suhr, 1992), investigating how symbolic cues operate within anthropomorphic design frameworks offers valuable insights into the socioemotional dynamics of human–AI interaction and informs the development of supportive chatbot systems.

2.2. CASA as theoretical framework

In human–computer interaction and human–machine communication research, the CASA paradigm (Nass et al., 1994) is a widely accepted framework that explains users' responses to human-like technologies. Social cues displayed by a computer can elicit social responses from people (Nass & Moon, 2000). At the same time, empirical findings on CASA are inconsistent. Some studies still reveal apparent CASA effects, such as in voice-activated navigation systems (Nass et al., 2005) and Alexa reciprocity interactions (Schneider, 2020), whereas others report null findings, including text-based chatbots (Tourangeau et al., 2003) and gendered robots (Neuteboom & de Graaf, 2021). Heyselaar's (2023) direct replication of the original CASA experiment further showed that people no longer display automatic social responses to desktop computers, suggesting that the presence or absence of CASA effects may depend on a novelty effect. That is, newer or socially unfamiliar technologies may elicit stronger CASA effects than routine, familiar ones. This variability has prompted scholars to reconceptualize CASA as more flexible and context-dependent.

Recent HMC scholars have suggested that the CASA paradigm has evolved from a narrow cue–reaction account into a flexible, context-sensitive framework suitable for relational and affective human–AI encounters. Specifically, as human behavior, technological capabilities, and patterns of human–machine interaction have substantially changed, Gambino et al. (2020) extended CASA by arguing that people mindlessly apply human–media social scripts rather than human–human ones, which better capture users' sophisticated and learned interactions with modern AI agents. Lombard and Xu (2021) further structurally extended CASA into the media are social actors (MASA) framework, distinguishing primary/secondary cues, integrating dual-path anthropomorphism, and highlighting contextual and individual moderating factors. Similarly, van der Goot and Etzrod (2023) extended CASA at a conceptual level by situating it within the media equation paradigm and contrasting it with media evocation processes that involve reflection, negotiation, and ontological boundary work.

Taken together, this contemporary view justifies applying CASA in the emotional support context examined here. And it motivates our focus on (a) perceived warmth and competence as core social-

cognitive judgments of supportive agents and (b) social presence as the activation mechanism through which identity and emoji cues can translate into relational outcomes.

2.3. Perceived warmth and perceived competence as core social perceptions of supportive chatbots

In the context of conversational AI, people may develop social perceptions of these digital agents, such as perceived warmth and perceived competence. Perceived warmth and competence are two universal dimensions of social perception used to identify if the other is a friend or foe (Fiske et al., 2007). These dimensions are central to the stereotype content model (SCM), which suggests that people intuitively assess others along these two axes in forming impressions. Recent chatbot studies have adopted this framework to examine how users perceive chatbot personalities and relational capacity (e.g., Belanche et al., 2021; Huang et al., 2021).

Scholars have found that people can perceive warmth and competence from machines (Bigman & Gray, 2018; Ho et al., 2018), and have explored the effect of an agent's emotional expression on people's perception of warmth and competence (DeMeur et al., 2011; Liu et al., 2021). For example, DeMeur et al. (2011) found that people perceived agents as warmer and more competent when emotions were expressed appropriately and plausibly, both verbally and nonverbally, rather than inappropriately or verbally alone. However, studies have also found inconsistencies in people's perceptions of competence and warmth. For example, Huang et al. (2021) found that using visual nonverbal cues could lead to higher perceptions of warmth but lower perceptions of competence for chat agents. In the business context, Kull et al. (2021) showed that a warm initial message from the chatbot could foster closer connections between consumers and the brand, compared to a less warm but more competent message. In the supportive chatbot context, Meng et al. (2023) further demonstrated that users' perception of a chatbot's competence after interaction could lead to cognitive reappraisal and stress reduction.

Despite the growing application of the SCM in chatbot research, existing studies often treat perceived warmth and competence as direct outcomes of interface cues, without fully addressing the cognitive mechanisms through which users activate these social schemas. According to studies following the CASA framework, people do not automatically apply social scripts to all machines under any circumstances; rather, such scripts are triggered only when the technology exhibits relational cues and is perceived as an autonomous source (Gambino et al., 2020). Building on this framework, we propose that perceived social presence may serve as an activating mechanism that enables users to develop warmth and competence judgments in human-chatbot interactions.

2.4. Social presence as an activating mechanism

Social presence has been defined differently depending on the type of cues it involves (Xu & Liao, 2020). In mediated communication, it refers to the perceived realness of the interaction partner, often involving feelings of intimacy and immediacy (Short et al., 1976). In human–chatbot interactions, it is widely conceptualized as the sense of "being with another" (Biocca et al., 2003). The principle of medium-as-social-actor presents that chatbots equipped with anthropomorphic cues are perceived by users as social entities, thereby leading users to generate responses similar to those in human-to-human interaction (Lombard & Ditton, 2006). This idea is supported by evidence that chatbots with human-like traits, including the use of emoji and human-like linguistic styles, can foster a sense of social presence for users (e.g., Araujo, 2018; Park & Sundar, 2015), resulting from a higher level of human-likeness (Beattie et al., 2020; Liu et al., 2018).

Social presence has been shown to promote emotional connection, trust, and perceived support, especially in human–machine contexts (Kim et al., 2013; So & Brush, 2008). These effects are not limited to emotional outcomes but extend to users' social judgments of agents. Prior work suggests that social presence may influence stereotype-based perceptions, such as warmth and competence, two core dimensions of social cognition (e.g., Dinh & Park, 2024; Van Doorn et al., 2017). However, previous studies remain mixed results: while some studies report positive associations between social presence

and both warmth and competence, others find that social presence affects perceived usefulness more than warmth (Merrill et al., 2022).

As noted earlier, certain design cues may boost warmth but fail to affect competence (Yu & Zhao, 2024), or even backfire depending on context (Huang et al., 2021). These inconsistencies suggest that design cues alone are insufficient to trigger stable stereotype perceptions unless users first experience social presence. This aligns with the idea that social presence functions as a perceptual gateway, bridging surface-level design and deeper psychological processing (Biocca et al., 2003; Lombard & Ditton, 2006). Building on this prior research, the current study aims to elucidate the underlying mechanism through which social presence influences perceived emotional support via perceived warmth and competence. Additionally, we hypothesize that social presence as an activating mechanism will have positive effects on users' social judgments toward the human agent and chatbot providing support.

2.5. Effects of emoji use

An emoji is defined as "a small digital image or icon" (Schneebeli, 2017) used to express emotions, ideas, or concepts in digital communication. Designed to enrich written text with personality, context, and humor, emojis can effectively convey emotional nuances that may be difficult to articulate through words alone (Bai et al., 2019). As their use has become increasingly prevalent in online communication, scholars have paid growing attention to how emojis influence emotional outcomes, such as more favorable impressions and evaluations, and improved mood (Coyle & Carmichael, 2019; Ganster et al., 2012). Emojis have also been examined for their capacity to trigger emotional shifts during interactions (Mostafavi & Porter, 2021).

Building on this line of research, recent studies have explored the integration of emojis into chatbot conversations, particularly as social cues that may shape users' perceptions and emotional responses. Fadhil et al. (2018) found that users felt more at ease discussing mental health issues with an emoji-using chatbot. Beattie et al. (2020) further found that emojis enhanced social attractiveness, computer-mediated communication competence, and credibility across both human and robotic communication contexts, underscoring their value as nonverbal social cues. Despite this promise, empirical findings on the emotional and relational impacts of emoji use in chatbot design remain mixed. Zlatkute et al. (2021) reported no significant effect of emojis on user trust and satisfaction, suggesting that such outcomes may depend on contextual or design factors.

In the present study, emoji use is conceptualized as part of the chatbot's linguistic style, which can prompt users to perceive machines as social actors (Fogg, 2003). Go and Sundar (2019) further found that anthropomorphic visual cues enhance users' sense of social presence, thereby fostering positive attitudes and emotional closeness. Building on these insights, this study applies the emoji-as-social-cue framework to the context of supportive chatbots, investigating whether different identity cues moderate the impact of emoji use on perceived emotional support.

2.6. Effects of identity cues

In technology-based communication, an identity cue refers to information indicating the identity of the communication partner (Go & Sundar, 2019). In studies comparing human and machine interlocutors, the identity cue is commonly manipulated as a disclosure of whether the agent is a chatbot or a human (Park et al., 2023). Self-claiming to be human on the part of the chatbot can enhance people's sense of social presence, leading to improved perceptions of competence and warmth toward the chatbot (Poinsot et al., 2022; Toader et al., 2019). However, there is concern that the use of human-related identity messages in chatbots may lead to confusion and a sense of deception, which may result in rejection (Park et al., 2023).

Previous research also shows contradictory findings on the effects of identity cues. Some studies found no substantial differences between human-to-human interaction and human-chatbot interactions regarding people's perceptions and disclosures (e.g., Ho et al., 2018). In line with the CASA paradigm (Nass et al., 1994), the emotional support received from chatbots can be perceived as being equivalent to that received from a human agent (Abendschein et al., 2021). However, there are also studies

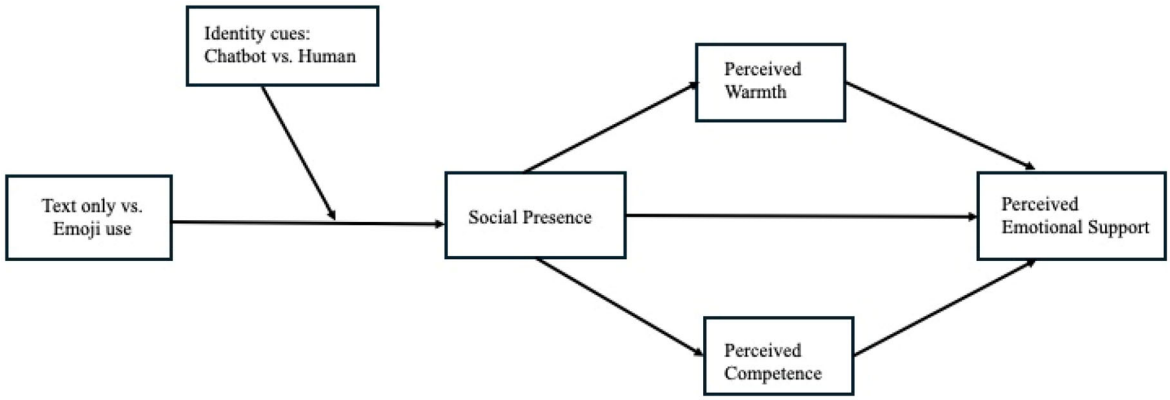

Figure 1. Research model.

showing that people attribute different judgments toward machines and humans. Medeiros et al. (2022) found that users reported greater stress reduction when they believed the support was provided by a human, while the least effective outcomes occurred when they knew it was computer-generated. This implies that the perceived identity of the support provider can significantly influence the effectiveness of emotional support, which may be potentially more powerful than the content of the message itself. Moreover, identity cues may activate common heuristics about machines, such as being cold and lacking empathy (Sundar & Kim, 2019), which may further weaken the impact of emotional cues like emojis conveyed by chatbots, limiting their ability to elicit perceived warmth and support.

Based on the discussion of related work above, three hypotheses and a research question are proposed as follows:

H1. Compared to text only, using emojis in the message will lead to a higher level of social presence.

RQ1. How does the identity cue (human vs. chatbot) moderate the influence of emoji use on social presence?

H2. The interaction between emoji use and identity cues will indirectly affect perceived emotional support through social presence.

H3. The interaction between emoji use and identity cues will indirectly affect perceived emotional support (a) through the social presence and perceived warmth in serial; and (b) through the social presence and perceived competence in serial.

The research model of this study is shown in Figure 1.

3. Method

3.1. Design and procedure

In this study, a 2 (identity cues: human vs. chatbot) by 2 (text only vs. text + emoji) online experiment was conducted using an interactive questionnaire. The questionnaire consisted of two parts: (a) an embedded chatbot interaction and (b) subsequent measures assessing participants' perceptions and responses. All participants were randomly assigned to one of the four conditions. Participants were told that they would have a conversation with a chatbot or a human agent before the questionnaire. The chatbots in this study were built using Sanuker, an existing chatbot development platform (http://www.sanuker.com/), which allows customization the conversation. Specifically, our chatbots can identify keywords from the user's typing and provide a response based on them. Sanuker was embedded in the online questionnaire. We used traditional Chinese in the experimental design and questionnaire to accommodate the language preference. In addition, we invited native speakers to refine the wording of the experimental material and questionnaire. After completing the interaction, participants responded to items measuring social presence, warmth, competence, and perceived emotional support.

To enhance the manipulation of the conversation partner identity, human avatars and names were used, and a message reading "the partner was typing…" was displayed in the human condition. A text alert was provided to remind different identity cues of the agent at the beginning of the chat. In addition, it was assumed that the chatbot would be able to provide more timely responses than the human partner, who, in the beginning, would take one minute longer to respond than the chatbot. In the chatbot identity condition, the text alert was "You will talk with a chatbot…" and the greeting would appear immediately. In the human identity condition, the text alert was "You will talk with a human partner, and please wait a minute, she/he is coming…" and then the greeting would appear after one minute.

In addition, the content of the greeting was different for the two identity cue conditions. In the human identity condition, the greeting was: "Hi, nice to meet you! My name is Hsi-Min Chen. I am a counselor at the Mental Health Institute. I am very happy to chat with you. You can share any of your thoughts and feelings with me." In the chatbot identity condition, the greeting content was: "Hi, nice to meet you! I am a mental health chatbot designed by Mental Health Institute. I am very happy to chat with you. You can share any of your thoughts and feelings with me." (Figure 2).

After the greeting, the participant was asked a question to self-disclose about stress issues. After the participant revealed these, the chat partner provided a text-only response or a text and emoji response, depending on the experimental condition. This completed one round of the conversation. The scripts of the text reply were the same for every condition. There were 11 rounds of the entire conversation (see Supplementary Appendix A for the full script, which includes both text-only and text + emoji conditions). To ensure that users completed the interaction with the chatbot, we assigned them a unique verification code at the end of the interaction, which they were asked to fill in at the beginning of the survey.

According to Beattie et al. (2020), the affective value, number, and placement of the emojis in the experiment can simulate comparable conversations in real life. The emojis we adopted in the scripts were based on the frequently used emojis that Hong Kong users use in similar chat conversations in WhatsApp. WhatsApp is the most widely-used instant communication application in Hong Kong, with nearly 85% of Hong Kong people using it in their daily lives (Datareportal, 2022).

We conducted three rounds of the pretest to ensure that the chatbot's features and stimulus material would work successfully. Through the sampling recruited from Qualtrics, we received 150 participants'

Figure 2. Examples of the identity cues across different conditions.

valid responses. The content of the script, the size and category of the emojis, and the wording of the questionnaire were modified based on responses from participants in the pretests. The samples in the pretests were not included in the formal experiment.

To control the influence of different genders of the human identity cues on users, we set up different genders of agents (female vs. male) in the human identity cues condition. Pretest analyses showed no significant gender effects, and we therefore collapsed the two gender versions into a single human identity cue condition in the final 2×2 design.

3.2. Participants and procedure

Participants were recruited through Qualtrics, a professional online panel provider widely recognized for its survey capabilities, during September and October 2022. Eligible participants were Hong Kong residents aged 18 years or above, with the age and gender distribution aligned with Hong Kong's general population. Respondents were invited through Qualtrics' proprietary panel system and completed the online questionnaire via a secure survey link. Participants received modest monetary compensation for their time upon successful completion of the survey.

As the study focused on the use of AI chatbots for emotional support, eligibility was limited to Hong Kong residents who had recently experienced depression, stress, or anxiety. A screening question was presented at the beginning (e.g., "Have you been feeling depressed, anxious, or stressed lately?"). Participants who answered "no" were not allowed to proceed to the subsequent experiment. Although the study was conducted in a mental health context, it did not specifically target individuals diagnosed with mental health disorders. Instead, it aimed to examine ordinary users' willingness to use an emotional-support chatbot when experiencing mild emotional distress.

To ensure participant safety and to rule out severe distress, we conducted a brief screening using the 9-item Depression Anxiety Stress Scale (DASS). The results showed no significant differences in DASS scores across the four experimental conditions (range = 4.22–4.43, $p > 0.10$), confirming that randomization was successful and that participants were not experiencing severe emotional distress. The study posed minimal risk to participants, and all respondents were informed that they could withdraw at any time without penalty. The research protocol was reviewed and approved by the Institutional Review Board (IRB) of the corresponding author's university.

We did *a priori* power analysis using G*Power 3.1 (Faul et al., 2009) to estimate the sample size. The results showed that 330 was the minimum sample size with a power of 0.8, alpha = 0.05 and effect size f = 0.20. A total of 637 individuals participated in the study, and the final valid sample consisted of 494 participants. Participants were excluded if they did not fill in the correct certifying code ($n = 15$). We conducted a thorough review of the chat records and excluded those that lacked valuable interactions ($n = 73$). This included participants who did not respond to the chatbot's questions or provided irrelevant answers. Additionally, we excluded 25 participants who failed the manipulation check and 30 participants with invalid responses. This group included 22 participants with unusually short or long completion times and 8 participants who displayed inconsistent answering patterns, such as selecting the same option throughout the survey. As a result, the final sample comprised 494 participants: 234 male (47.4%), 258 female (52.2%), and 2 participants identifying as other genders (0.4%).

3.3. Measurement

3.3.1. Social presence

A 7-point scale adopted from Park and Sundar (2015) were used to measure social presence (Cronbach's $\alpha = 0.87$). Examples of items were "the chatbot/human agent could tell how I felt," "the chatbot/human agent was sometimes influenced by my moods," and "the chatbot/human agent influenced the mood of our interaction."

3.3.2. Perceived competence and perceived warmth

Perceived competence was measured using 7-point Likert scale adopted from Fiske et al. (2007). The scale ranged from strongly disagree (1) to strongly agree (7). Five items were used to measure the

perceived competence (Cronbach's $\alpha = 0.91$) and 4 items were used to measure the perceived warmth (Cronbach's $\alpha = 0.88$). Examples of items were "As viewed by you, how competent do you think this chatbot/human agent is?," "As viewed by you, how tolerant do you think this chatbot/human agent is?" and "As viewed by you, how warm do you think this chatbot/human agent is?"

3.3.3. Perceived emotional support

Nine items adapted and revised from a previous study (Zimet et al., 1988) measured participants' perceived emotional support on a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree, Cronbach's $\alpha = 0.94$, M = 4.33, SD = 1.19). Examples of items from the scale were: "There is a human agent/chatbot who is around when I am in need." "I get the emotional help and support from the human agent/chatbot." And "The human agent/chatbot is a real source of comfort to me."

3.4. Manipulation check

Two questions were included to assess the manipulations of identity cues and emoji cues at the beginning of the survey. Following Go and Sundar (2019), participants were asked to identify the conversational partner's identity (1 = female human agent, 2 = male human agent, 3 = chatbot) and the message elements (1 = text only, 2 = text and emoji). The purpose of this manipulation check was to ensure that participants clearly recognized the assigned identity and the presence or absence of emojis. 25 participants who failed this manipulation check were excluded from the final analyses.

4. Results

H1 posited the effect of emoji use in the message on social presence. An independent *t*-test showed that participants perceived more social presence when the communication partner used emojis ($M = 4.48$, SD = 0.96) than when there is only text ($M = 4.07$, SD = 1.10) in the message, $t = -4.45$, $p < 0.05$. Thus, H1 was supported.

To answer the RQ, we conducted a 2 (emoji use vs. text only) × 2 (identity cues: chatbot identity vs. human identity) ANCOVA analysis on the social presence. The results showed that the interaction between emoji use and identity cues significantly affected the social presence ($F[1, 490] = 46.95$, $p = 0.009$, partial $\eta^2 = 0.01$). There is an interaction effect between emoji use and identity cues on social presence. We then did the simple effects analysis using the Fisher's least significant difference (LSD) method. The results showed that for the human identity cue condition, using emoji ($M = 4.62$, SD = 0.09) led to a significantly higher level of the social presence than the condition of the text ($M = 4.00$, SD = 0.09), $t = 5.11$, $p < 0.01$. For the chatbot identity cue condition, the difference between emoji ($M = 4.30$, SD= 0.10) and text-only ($M = 4.17$, SD = 0.10) was not significant in the social presence ($p = 0.36$). (See Figure 3. for the interaction plots.)

To test H2 and H3, we further customized a moderated serial mediation model following Hayes' (2018) PROCESS instructions. The indirect effect was tested using 5000 bias-corrected bootstrap samples with 95% confidence intervals. We set emoji use as the independent variable, identity cues as the moderator, social presence, perceived warmth and perceived competence as mediators and perceived emotional support as the dependent variable.

For H2, the moderated mediation effect regarding social presence was significant (index = 0.15, SE = 0.07, 95% CI [0.04,0.30]). The results showed that the indirect effects of emoji use through social presence were different between the chatbot condition and the human condition. The indirect effect of emoji use through social presence was not significant in the chatbot identity cue, while it was significant in the human identity cue ($b = 0.20$, SE = 0.053, 95% CI = [0.11, 0.30]). Therefore, H2 was supported.

For H3, the moderated mediation index was significant on perceived emotional support through the social presence and perceived competence (index = 0.22, SE =0.09, 95% CI [0.06, 0.41]), and through the social presence and perceived warmth (index =0.04, SE= 0.02, 95% CI [0.01, 0.09]). For the chatbot identity cue condition, the indirect effect of emoji use on perceived emotional support through the social presence and perceived competence was not significant. In addition, the indirect effect of emoji

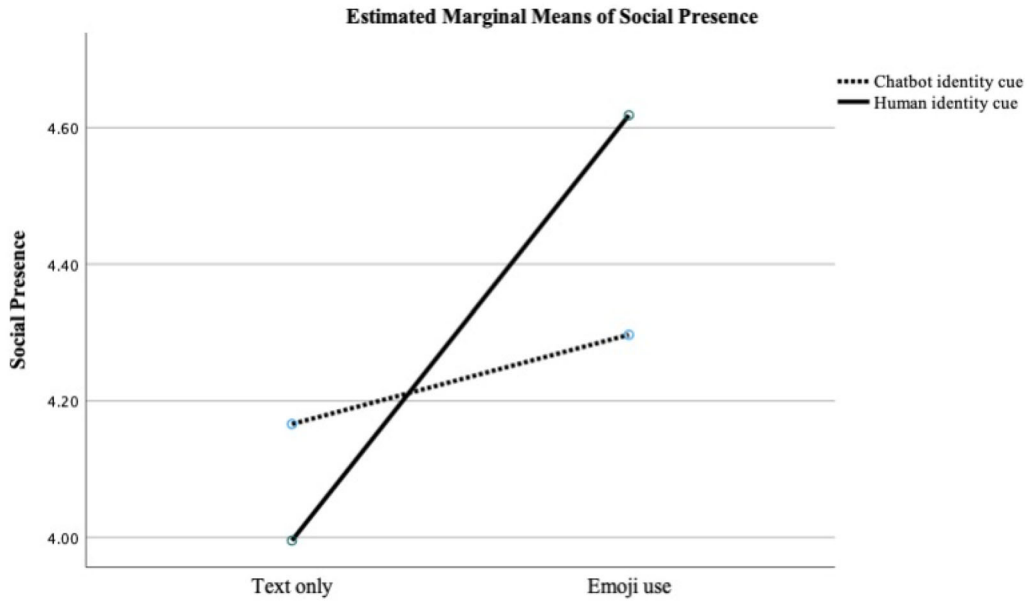

Figure 3. Interaction between emoji use and identity cues on social presence.

Table 1. Indirect effects of emoji use on perceived emotional support via (a) social presence and perceived competence in serial, (b) social presence and perceived warmth under human identity cue condition.

		Indirect effect 95% confidence intervals	
	Indirect effect bootstrap	Confidence intervals	
Mediation path	Estimate (b)	LL	UL
a) Emoji use → Social presence → Perceived competence → Perceived emotional support	0.28(0.06)	0.05	0.41
b) Emoji use → Social presence → Perceived warmth → Perceived emotional support	0.05(0.02)	0.02	0.09

use on perceived emotional support through the social presence and perceived warmth was not significant. For the human identity cue condition, the indirect effect of emoji use on perceived emotional support through social presence and perceived competence was significant ($b = 0.28$, SE = 0.06, 95% CI = [0.17, 0.41]).

Furthermore, the indirect effect of emoji use on perceived emotional support through social presence and perceived warmth was significant ($b = 0.05$, SE = 0.02, 95% CI = [0.02, 0.09]). Thus, H3a and H3b were supported. Overall, the results showed that identity cues moderated the impacts of emoji use on perceived emotional support through the social presence. Only in the human condition could using emojis in the message increase the level of social presence of the participants and enhance their perceived warmth and perceived competence, respectively, further increasing their perceived emotional support (Table 1).

5. Discussion and implications

5.1. General discussion

This study examined how emoji use and identity cues in chatbot design jointly shape users' perceptions of emotional support. Our findings illustrate both the process and boundary conditions under which empathic chatbots enhance users' support experiences. First, chatbots foster a sense of social presence by integrating subtle human-like message features, such as emojis rather than plain text. Second, identity cues serve as a boundary condition: emoji use enhanced social presence and perceived support only when the chatbot was presented with a human-like identity. Third, we identify social presence as the key psychological mechanism driving supportive communication between humans and chatbots.

Specifically, emoji use and identity cues influenced perceived emotional support through two serial paths: (a) social presence and perceived competence, and (b) social presence and perceived warmth.

These results reveal that conversational cues like emojis are not inherently effective, they depend on the presence of anthropomorphic identity cues. This layered interaction supports prior arguments (Tsai & Chuan, 2024) that identity and dialogue features must work together to drive positive user responses. In the human identity condition, people may apply human–human schemata to explicate the messages, enhancing perceived sincerity and sensitivity due to emojis acting as nonverbal cues that influence tone and meaning (Ganster et al., 2012; Meng & Dai, 2021). So, people are more likely to perceive the emojis from human identity cues as an attempt to clarify, contextualize, or interpret the messages in depth (Prada et al., 2018).

Importantly, this study also underscores the role of perceived competence in emotional support contexts. While earlier work (e.g., Yu & Zhao, 2024) found emoji use increased warmth but not competence in general chatbot interactions, our findings suggest that in emotionally supportive settings, users care about both the chatbot's emotional sensitivity and its perceived ability to understand and respond effectively. However, our findings also highlight a clear drawback: emotional messaging strategies, including emojis are largely ineffective when chatbots lack human identity cues. This may be explained by the AI attribution effect (Yin et al., 2024), whereby users discount emotional messages upon recognizing AI generation, triggering skepticism. Moreover, the mismatch between a machine-like identity and emotional cues may further create expectation violations (Rheu et al., 2024), inducing cognitive dissonance and reinforcing the stereotype that machines are cold and unfeeling (Sundar & Kim, 2019; Prahl & Van Swol, 2021). Extending this logic, such reactions may also indicate task-dependent algorithm aversion (Castelo et al., 2019), as emotional-support tasks inherently challenge users' belief in human distinctiveness, the notion that empathy and affective understanding are uniquely human capacities. When an AI agent expresses emotion, it may therefore evoke discomfort or disbelief, causing users to downplay its social and emotional intentions. This perspective advances HMC theory by elucidating how cognitive inconsistency, social attribution, and perceived threats to human uniqueness jointly shape users' interpretation of affective cues in AI-mediated interactions.

5.2. Theoretical and practical implications

Our research has several theoretical contributions. First, this study strengthens and extends the contemporary evolution of the CASA paradigm by delineating a perception–cognition–behavior pathway in supportive human–AI communication. Consistent with recent expansions of CASA into a flexible, context-sensitive framework (e.g., Gambino et al., 2020; Lombard & Xu, 2021), our findings show that human-like social cues (emoji use and identity cues) enhance perceived social presence, which subsequently shapes evaluations of warmth and competence and ultimately influences perceived emotional support. In doing so, the study specifies a mechanistic extension of CASA, clarifying how social presence functions as an activating and serial mediating process connecting cues to relational outcomes. Moreover, the results identify a boundary condition within CASA: the effectiveness of anthropomorphic cues such as emojis depends on the alignment with human identity cues. This insight aligns with Beattie and High (2022), suggesting that cumulative, coherent human-like cues are necessary to activate social scripts in supportive HMC.

Second, it advances chatbot design literature by highlighting the context-dependent nature of social cues, showing that emotional support emerges from the interplay of emoji use and identity cues. This suggests designers should leverage multi-layered cues to activate human–human schemata (Meng & Dai, 2021), as supported by Feine et al. (2019) on creating natural interactions.

Third, this research contributes to cross-cultural theorizing by conducting the study in Hong Kong, a representative high-context communication culture where users are more attuned to nuanced and layered social cues. This setting underscores the importance of cue coherence and relational expectations across cultures, offering a valuable extension of supportive chatbots research beyond Western contexts.

Fourth, the present findings contribute to recent developments in HMC that explore how people form relational, affective, and moral evaluations of AI agents (Guzman & Lewis, 2020). By demonstrating that emotional cues such as emojis enhance perceived warmth, competence, and emotional support

only under human identity framing, this study offers empirical insight into how social presence and identity cues jointly shape users' relational perceptions of chatbots. These results complement emerging HMC discussions about authenticity and relationality (Etzrodt et al., 2024), suggesting that users' emotional responses depend less on the agent's surface human-likeness than on the coherence between its expressive style and presented identity. In doing so, the study situates the CASA paradigm within broader HMC perspectives that emphasize contextual, interpretive, and relational responses rather than uniformly mindless reactions.

This study also provides ethical and practical implications for designing emotionally supportive chatbots. The observed asymmetry highlights an ethical dilemma in HMC—while emotional expressiveness helps chatbots appear warmer and more supportive, it also risks obscuring their nonhuman identity. As Etzrodt et al. (2024) note, authenticity in HMC requires acknowledging both the social and technical dimensions of machine communication. Designers should therefore pursue strategic authenticity: calibrating emotional expression to a clearly disclosed nonhuman identity and maintaining coherence between linguistic style and disclosed status. Although the present study used a pre-scripted chatbot, its findings inform how this principle applies to LLM-based supportive communication. The expressive fluency of LLMs can intensify the "hallucinatory danger" of fostering false impressions of empathy or care (Bisconti Lucidi & Nardi, 2018; Fox & Gambino, 2021). Without careful calibration, such systems may heighten the ethical risks identified by Coghlan et al. (2023), including misplaced trust, inappropriate self-disclosure, and failures in crisis response. As mental health chatbots become more prevalent, particularly among younger users, ethical deployment must pair accessibility with transparency, embedding contextual sensitivity and explicit boundaries of capability.

5.3. Limitations and future research

Several limitations of the current study must be acknowledged. First, our experimental chatbot relied on pre-scripted dialogues to ensure precise manipulation of identity and social cues. While this approach strengthens internal validity, it limits ecological validity relative to real-world conversational agents whose language, tone, and identity expressions evolve dynamically. Future research should extend these findings to more naturalistic contexts, particularly with LLM-based chatbots (e.g., ChatGPT, DeepSeek), where identity cues may emerge implicitly through linguistic style rather than through explicit framing. Examining how users interpret and respond to emergent versus designed identity signals in such systems would deepen the understanding of authenticity and trust in evolving HMC contexts. Additionally, longitudinal research could assess whether short-term perceptions of warmth and support translate into sustained engagement or unintended emotional dependence during prolonged real-world use.

Second, this study focused on the effects of emoji use in supportive chatbots within Hong Kong. Acceptance of emojis and expressive cues varies across cultures (Konrad et al., 2020), which may limit the generalizability of our findings. Future research could conduct cross-cultural comparisons to examine whether the role of emotional cues and identity framing differs across sociocultural settings.

Third, users' prior experience and literacy about AI applications can profoundly influence their perceptions of and how they behave in relation to chatbots for emotional support. It would also be worthwhile in future studies to investigate the moderating effects of individual characteristics in human-chatbot interactions.

Generative AI disclosure

This manuscript was proofread using ChatGPT 4.0 to improve grammar, clarity, and overall readability. The authors reviewed and edited the content and take full responsibility for the final version of the publication.

Author contributions

CRediT: **Yu-li Liu**: Conceptualization, Funding acquisition, Supervision, Writing – review & editing; **Chuling Song**: Conceptualization, Investigation, Methodology, Project administration, Writing – original draft, Writing –

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Data availability statement

The data supporting this study's findings are available from the corresponding author upon reasonable request.

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